

Longest Palindrome

Time limit: 1.00 s

Memory limit: 512 MB

Given a string, your task is to determine the longest palindromic substring of the string. For example, the longest palindrome in aybabbtu is bab.

Input

The only input line contains a string of length n . Each character is one of a–z.

Output

Print the longest palindrome in the string. If there are several solutions, you may print any of them.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
aybabbtu	bab